Zaid Hanif

(304) 276-8251 | zah00005@mix.wvu.edu | zaidhanif.dev | linkedin.com/in/zaidhanif | github.com/zaidhanif1

Education

West Virginia University — Morgantown, WV
Bachelor of Science in Computer Science (GPA: 3.63/4.0) — *Aug. 2023 – May 2027*

Universitat Pompeu Fabra — Barcelona, Spain
Study Abroad – Computer Science & Language Coursework — *Jan. 2025 – Apr. 2025*

Experience

Software Engineering Intern — May 2026 – July 2026
Southwest Research Institute (SwRI) — *San Antonio, TX*

- Built Python backend features for an internal AI platform, developing a retrieval pipeline and FastAPI service that transformed technical documents into context-grounded answers via semantic embeddings and top-k vector search.
- Architected persistent search layer using SQLAlchemy, SQLite, and sqlite-vec; added checksum deduplication to skip redundant ingestion, cutting startup time by 7x.
- Refactored ingestion, storage, retrieval, and generation into a modular Query API using Python dataclasses, typed vector serialization, and structured JSON logging.
- Containerized a FastAPI service exposing /upload and /query endpoints, with Pydantic configuration and internal Ollama server integration for local LLM inference.

Projects

SlotSaver | *React, Next.js, TypeScript, FastAPI, PostgreSQL, AWS (EC2/RDS)* — Pittsburgh, PA
SteelHacks XIII – PNC Compound Track Winner — *Sep. 2026*

- Built a full-stack appointment cancellation and rebooking platform with a Next.js/React frontend and Python/FastAPI backend, matching open appointments with customers based on scheduling preferences and availability.
- Built an end-to-end workflow that filters eligible customers, uses NVIDIA Nemotron to rank candidates, and triggers ElevenLabs voice agents with Twilio for outbound calls.
- Engineered an atomic rebooking workflow with **PostgreSQL** transactions to eliminate race conditions, ensuring **100% data accuracy** when resolving stale states and concurrent appointment timeouts.
- Deployed application components across highly available **AWS EC2** and **RDS** infrastructure instances to ensure platform scalability and fault tolerance.

Momentum Fitness Tracker | *React, TypeScript, Express, MySQL*

- Built workout tracker with React and TypeScript frontend and Express REST API connected to a MySQL database over SSL.
- Implemented token-based authentication with short-lived access tokens and 90-day refresh tokens, tracking and revoking active sessions server-side.
- Wrote streak algorithm converting workout timestamps into unique calendar dates to count consecutive activity days.
- Led an 8-member software engineering team, translating product requirements into scoped tasks and coordinating Agile sprints.

MPX Operating System | *C, x86 Assembly*

- Developed a bare-metal x86 kernel, implementing the GDT, IDT, PIC, virtual memory paging, and physical memory management in a freestanding C environment.
- Implemented cooperative multitasking through custom x86 assembly ISR saving and restoring 16 registers per context switch, prioritizing scheduling to maintain kernel process **high availability**.
- Built first-fit dynamic heap allocator using Memory Control Blocks with block splitting and adjacent merging to minimize fragmentation.
- Engineered interrupt-driven serial I/O driver for Intel 8250 UART, managing Device and I/O Control Blocks with ring-buffered input.

Technical Skills

Languages: Go, Kotlin, Python, TypeScript, JavaScript, SQL, C++, C, Java, Assembly
Frameworks: React, FastAPI, Next.js, SQLAlchemy, Express.js, Pydantic, LangChain, Tailwind CSS
Tools/Databases: AWS (EC2/RDS), PostgreSQL, MySQL, SQLite, Docker, Linux, sqlite-vec, Git, GitHub, GitLab, Ollama
Spoken Languages: English (Native), Arabic (Bilingual), Spanish (Intermediate)

Leadership / Student Organizations

Society of Hispanic Professional Engineers — Aug. 2025 – Present
Treasurer — *Morgantown, WV*